

BidWise

Governance product engineering quality and support documentation

BidWise is an evidence-led tender compliance review prototype. This pack gives executive sponsors, technical contributors and procurement users a single working baseline for delivery, review and handoff.

Document control	Baseline
Document owner	Manju, Project Lead and document controller
Primary audience	SIH evaluators, project sponsors, developers and procurement officers
Central workspace	Notion workspace titled BidWise Project Hub
Version	1.0 baseline
Document date	22 September 2026
Approval status	Ready for technical and procurement review

Manju maintains the controlled copy, records revision history and issues approval requests. The Technical Reviewer approves system and security content. The Procurement Reviewer approves rule, workflow and evidence language. The project mentor or sponsor provides final release approval before external submission.Ownership and approval

Abstract

BidWise supports first-level tender compliance review by extracting defined requirements from a tender PDF, comparing those requirements with bidder documents and presenting evidence references, status labels and risk flags for a procurement officer. The current prototype uses text extraction with a small rules library for network equipment clauses. It does not make an award decision, validate legal eligibility, or replace the authorised officer.

The addressable context is material. Government e-Marketplace recorded ₹5.4 lakh crore in Gross Merchandise Value in FY 2024-25, according to the Press Information Bureau. The same release describes a national digital procurement system that serves government buyers and sellers across goods and services [R1]. A March 2025 PIB release reported more than 1.6 lakh government buyers, more than 22 lakh sellers and service providers, and cumulative public savings exceeding ₹1.15 lakh crore [R3].

This pack defines the business case, operating controls, build scope, requirements, quality gates and long-term support approach for a credible pilot. It distinguishes current prototype functionality from planned product capabilities so that reviewers can evaluate the work on its actual merits.

Contents

- 1. Governance and document control
- 2. Project charter
- 3. Scope statement
- 4. Timeline budget and resources
- 5. RAID log and communication plan
- 6. Product and business requirements
- 7. Architecture and design specification
- 8. Standard operating procedures
- 9. Quality assurance and testing plan
- 10. User manual and release notes
- 11. Post mortem and maintenance support
- 12. References

Governance and Document Control

The documentation system has one owner and one source of truth. The team will keep all controlled documents in the BidWise Project Hub in Notion. Each document page carries a version, owner, status and review date. Supporting files, figures and approved exports are attached to the same Notion page or its linked repository record.

Control	Operating rule	Owner
Target audience	Executive sections explain business value and delivery risk. Technical sections specify implementation and tests. User sections use step-by-step instructions.	Manju
Document ownership	Manju maintains every controlled document until a pilot organisation assigns named operational owners.	Manju
Storage location	Notion: BidWise Project Hub. It contains charter, decision log, approval register, requirements, releases and reference links.	Manju
Version control	Use v0.x for drafts, v0.9 for review, v1.0 for approved baselines and v1.x for controlled minor revisions. Record date, author, change summary and approver.	Manju
Review and approval	Technical Reviewer signs off architecture, security and QA. Procurement Reviewer signs off requirements and evidence workflow. Manju records final approval and distribution.	Manju

Revision Register

Version	Date	Author	Change summary	Approval
1.0	22 Sep 2026	Manju	Initial integrated project documentation baseline	Pending technical and procurement review

Project Charter

Create a secure, explainable first-level review workspace that helps procurement teams find tender requirements, compare bidder evidence and direct attention to missing or uncertain documentation. Purpose

Charter element	Definition
Problem	Manual bid verification requires reviewers to read multiple PDFs, reconcile requirements and retain evidence for audit. The work is repetitive and easy to document inconsistently.
Goal	Produce a review-ready evidence register for scoped tender requirements while retaining officer control over the final decision.
Business value	Reduce document-search effort, apply the same first-level checks to each bidder and make evidence easier to retrieve during review or audit.
Success measure	For supported clauses, the system returns a status, reason and document-page reference. The officer can export or review the evidence register without reopening every source file.
Project authority	Manju is authorised to manage the prototype backlog, documentation baseline and internal release. Any live deployment requires sponsor, security and procurement approval.
Key stakeholders	Procurement officers, technical evaluators, bidder-facing support teams, IT security, system integrators and project evaluators.

Market Evidence

Official GeM reporting shows sustained growth in digital public procurement. The historical figure below appears in a PIB publication and uses GeM Transactions and Finance data. It shows FY17 to FY23 growth in reported GMV, ending at ₹2,01,113 crore in FY23 [R4].

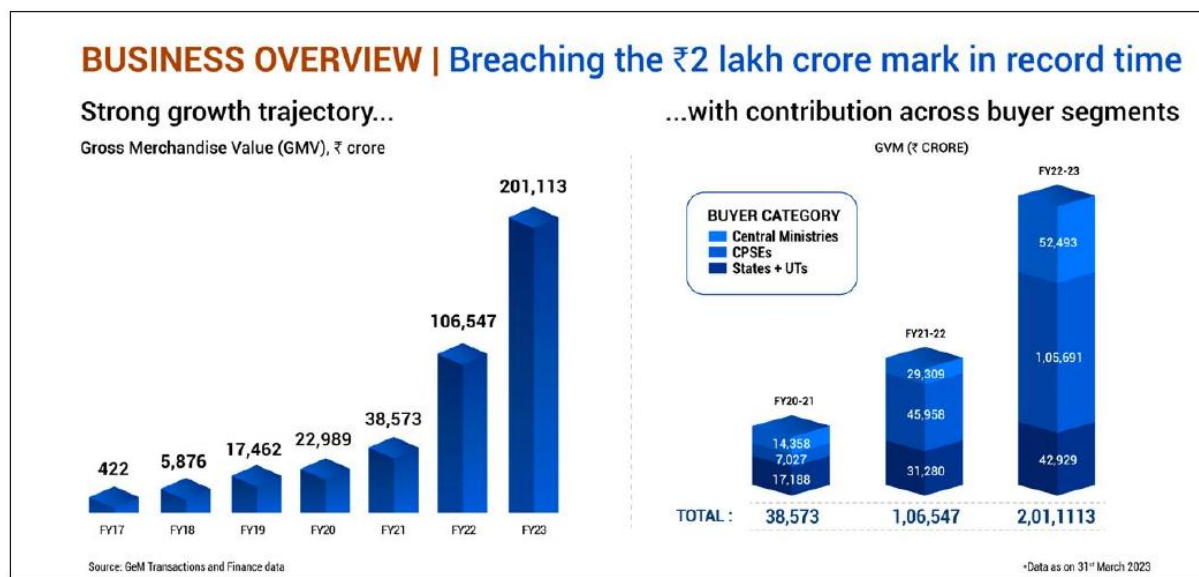


Figure 1. Official GeM Growth Journey chart. Source: PIB publication using GeM Transactions and Finance data, dated 31 March 2023 [R4].

Scope Statement

The scope separates the demonstrable prototype from a production procurement platform. This prevents reviewers from assuming that planned controls are already available.

Included in prototype	Excluded or planned later
PDF upload for one tender and one or more bidder documents	Live GeM integration and direct access to public procurement data
Text extraction from machine-readable PDF pages using pypdf	OCR for scanned PDFs and image-based tender pages
Rules for 24-port managed Gigabit switches, Wi-Fi 6 access points, warranty duration and onsite technical support	General clause understanding across every commodity, service and legal condition
Compliance score, missing-document list, risk list and evidence-page references	Automated tender award, bidder exclusion or legal eligibility decision
FastAPI backend, browser workspace, reference cases and downloadable review report	Production identity management, role-based access control, encryption key management and operational monitoring
Human review boundary in UI and documentation	Digital signature validation, fraud detection and official document authenticity verification

Scope Change Control

Any request for a new tender domain, source-system integration, automated decision or new data class enters the Notion change register. Manju estimates impact on rule coverage, security, test evidence and timeline. The Technical Reviewer and Procurement Reviewer approve the change before implementation starts.

Timeline and Milestones

The baseline uses a 12-week prototype plan. The timing is a delivery assumption, not a commitment from a procuring authority. Dependencies are shown through the sequence of shaded periods.

Workstream	W1	W2	W3	W4
Charter requirements and rules				
UX evidence register and reference cases				
Backend extraction and rule checks				
Frontend workspace and report export				
QA security review and defect resolution				
Pilot walkthrough and documentation				
Release readiness and submission				

Figure 2. Prototype delivery baseline. Shaded cells show planned work periods and dependency sequence.

Milestone Exit Criteria

Milestone	Exit criterion	Accountable owner
Requirements baseline	Scope and initial rule catalogue approved	Manju and Procurement Reviewer
Build complete	Supported PDFs return status, reason and evidence page	Technical Reviewer
Quality gate	Priority-one defects closed and test evidence reviewed	Manju and Technical Reviewer
Release ready	Documentation, reference links and user walkthrough complete	Manju

Budget and Resource Plan

The following is a planning estimate for a 12-week student prototype. It is not a quote, commitment or approved procurement budget. Before spending, Manju will replace these estimates with supplier quotes and sponsor approvals in the Notion budget page.

Category	Planning assumption	Estimated cost INR	Control
Cloud and hosting	Development hosting, test storage and API access	12,000	Monthly usage cap and review
Document processing	PDF test corpus, OCR evaluation credits and security scanning	8,000	Use synthetic or approved test files only
Security and domain review	External review time or specialist consultation	10,000	Written review record
Demo and user testing	Pilot preparation, materials and accessibility review	6,000	Session notes and consent
Contingency	Unplanned software or validation costs	14,000	Project Lead approval
Total	12-week planning baseline	50,000	Sponsor approval before spend

Resource Roles

Role	Core responsibility	Named owner
Project Lead	Backlog, documentation, delivery decisions and approvals	Manju
Frontend contributor	Workspace, document reader and evidence register interface	Team assignment in Notion
Backend contributor	FastAPI endpoints, PDF text extraction and rules	Team assignment in Notion
QA and domain reviewer	Test evidence, policy alignment and defect review	Technical and Procurement Reviewers

RAID Log

The RAID log is the central control record for delivery uncertainty. Manju updates it weekly, escalates high-priority items within one business day and records the decision or mitigation in Notion.

Type	Item	Impact	Response	Owner	Status
Risk	Scanned PDFs may not contain extractable text	High	Add OCR pipeline and confidence review before live pilot	Technical Reviewer	Open
Risk	Rule-based matching may miss valid wording variants	High	Use officer review queue and collect false-positive examples	Manju	Open
Risk	Prototype CORS policy permits any origin	High	Restrict origins and add authentication before any live deployment	Technical Reviewer	Open
Assumption	Pilot tender documents are non-sensitive or approved for use	High	Obtain written data-use confirmation before upload	Manju	Open
Assumption	Initial pilot focuses on network equipment requirements	Medium	Lock rule library and test corpus for the pilot	Procurement Reviewer	Open
Issue	Current backend lacks structured error messages for damaged PDFs	Medium	Add file validation and safe error responses	Backend contributor	Open
Dependency	Final workflow review requires procurement-domain input	High	Schedule formal review before release gate	Manju	Open
Dependency	Production controls require hosting and identity decisions	High	Select hosting model and security owner during pilot planning	Sponsor	Open

Communication Plan

Audience	Information	Cadence	Channel	Owner
Project team	Backlog, defects, design decisions and blockers	Weekly	Notion weekly update and stand-up notes	Manju
Technical Reviewer	Architecture changes, security gaps and QA evidence	At each release candidate	Notion review task and recorded sign-off	Manju
Procurement Reviewer	Rule coverage, evidence labels and human decision boundary	At requirements baseline and pilot	Walkthrough and approval log	Manju
Sponsor or mentor	Milestones, risks, budget changes and final readiness	Biweekly	One-page status update	Manju
End users	Release notes, user guide and support route	At release	Workspace release page	Manju

Product and Business Requirements

Give a procurement officer one evidence-led workspace for a repeatable first review. The officer must be able to understand why each status appears and retrieve the supporting document page.

Users and Needs

User	Need	Outcome
Procurement officer	Review requirements and missing proof without reopening every PDF	A clear evidence register and review queue
Technical evaluator	Check supported specifications against bidder submissions	Requirement-level status, reason and page reference
Project administrator	Configure the scope and retain decision evidence	Rules, logs, releases and documentation in one governed workspace
Bidder support team	Understand document expectations before submission	Consistent checklist guidance and feedback categories

Functional Requirements

ID	Requirement	Priority	Acceptance criterion
FR 01	Upload one tender PDF and one or more bidder PDFs	Must	The interface blocks non-PDF files and shows attached filenames
FR 02	Extract text from machine-readable PDF pages	Must	The API returns text for readable pages and retains source filename
FR 03	Detect supported network clauses	Must	The rule library detects 24-port switches, Wi-Fi 6, warranty duration and onsite support when the defined text appears
FR 04	Compare bidder text against detected clauses	Must	Each detected clause returns compliant, non-compliant or review status with a reason
FR 05	Record evidence page	Must	A matching page number and source excerpt appear when matching text is found
FR 06	Calculate compliance percentage	Must	Percentage equals compliant checks divided by total detected checks multiplied by 100 and rounded to two decimals
FR 07	Present officer review results	Must	Dashboard shows status counts, filters and evidence register
FR 08	Export assessment report	Should	User downloads the result table with score, status, reason and evidence

Non Functional Requirements

Area	Requirement	Prototype status
Security	Use HTTPS, restricted origins, authenticated roles and encrypted storage before live deployment	Target production control
Privacy	Do not retain tender or bidder files beyond the approved assessment retention period	Policy required before pilot
Explainability	Every status must show reason and evidence or state evidence was not found	Implemented for scoped rules
Reliability	Reject malformed documents safely and provide a clear message	Known gap to close
Performance	Process a standard machine-readable tender and bidder pack within the agreed pilot service target	Target to benchmark
Accessibility	Keyboard navigation, readable contrast and clear status labels	Design target and QA check

Architecture and Design Specification

The current architecture uses a static browser interface and a FastAPI service. The browser sends PDF files to a local endpoint. The service reads machine-readable PDFs through pypdf, identifies a small set of supported clauses with text and regular-expression rules, compares bidder text and returns a structured assessment. The browser renders the evidence-led review result.

Current Technical Flow

Tender	PDF	PDF	extraction	Rules	engine	Evidence	register	Officer	review
Source clauses		pypdf text		Scoped checks		Page links		Human decision	

Figure 3. Current BidWise processing flow. The officer performs the final decision outside the automated rules engine.

Component Specification

Component	Current behaviour	Production direction
Browser workspace	Provides reference cases, document attachments, processing states, result filters and report export	Add authenticated workspace, task history and accessible evidence preview
FastAPI service	Exposes PDF extraction, requirement extraction and tender-plus-bidder assessment endpoints	Add authentication, rate limits, validation, structured errors and observability
PDF extraction	Uses pypdf to retrieve embedded text from readable PDF pages	Add OCR with page confidence when scan quality requires it
Rules engine	Detects four scoped network requirements using text and regular-expression matching	Add configurable clause catalogue, synonyms, thresholds and versioned policy rules
Evidence mapping	Returns page number and matching text where a supported rule is found	Add immutable evidence snapshots, citation anchors and reviewer annotations

Security Design

The current implementation is a local prototype. It has no user authentication and its CORS policy permits all origins. Therefore it must not process confidential procurement records in its present state. A production deployment requires HTTPS, named accounts, role-based access, restricted origins, encryption in transit and at rest, retention controls, audit logs, antivirus scanning and a documented incident response process.

Standard Operating Procedures

Assessment Preparation

Step	Action	Control evidence
1	Confirm the tender and bidder documents are approved for use and do not contain restricted data outside the agreed environment.	Notion assessment record
2	Check that each file is a machine-readable PDF and label it with tender reference, bidder and document type.	File register
3	Open the BidWise workspace and attach the tender first, then one or more bidder documents.	Assessment timestamp
4	Run the assessment and review every action or review status before discussing any result.	Evidence register
5	Record officer comments and the final decision outside the automated result. Export the report if the review requires a static record.	Decision log and export

Rule Change Procedure

Step	Action	Approval
1	Describe the tender clause, acceptable evidence and rule outcome in the rule-change request.	Procurement Reviewer
2	Create positive, negative and ambiguous test excerpts.	Technical Reviewer
3	Implement the rule with a version identifier and evidence logic.	Backend contributor
4	Run regression tests against existing supported clauses.	QA reviewer
5	Record release note, approver and effective date in Notion.	Manju

Exception Handling

If the system cannot extract text, finds no supported requirement, or produces a result that conflicts with the document, the officer must mark the assessment as manual review. The team records the case without uploading confidential content to uncontrolled channels. The Technical Reviewer classifies the issue as data quality, rule coverage, extraction defect or interface defect.

Quality Assurance and Testing Plan

A release meets the quality gate when all must-have acceptance criteria pass, priority-one defects are closed, each supported rule has positive and negative test evidence, the human review boundary is visible in the interface, and the technical and procurement reviewers approve the release record. Definition of done

ID	Test case	Expected result	Status
TC 01	Upload supported tender PDF containing 24-port clause	Requirement list includes 24-port managed Gigabit switches	Required
TC 02	Upload tender PDF containing Wi-Fi 6 clause	Requirement list includes Dual-band Wi-Fi 6 access points	Required
TC 03	Upload tender with minimum warranty duration	Rule captures the stated year count	Required
TC 04	Bidder includes exact required switch clause	Status is compliant and evidence page is recorded	Required
TC 05	Bidder uses 802.11ax for Wi-Fi 6	Status is compliant under the defined equivalent-text rule	Required
TC 06	Bidder warranty is below tender minimum	Status is non-compliant with a clear reason	Required
TC 07	Bidder lacks onsite technical support wording	Status is non-compliant and appears in risk list	Required
TC 08	Tender contains no supported clause	API returns no requirements found and UI directs user to review scope	Required
TC 09	Upload a non-PDF file	Interface rejects the file before assessment	Required
TC 10	Use malformed or password-protected PDF	Production target: safe validation error with no stack trace	Open gap
TC 11	Run all four reference cases	Each case opens document previews and a coherent status table	Required
TC 12	Export report	Export contains the current score, requirements, reason and evidence fields	Required
TC 13	Keyboard navigation	Controls, filters and links can be reached and activated by keyboard	Required
TC 14	CORS and access control review	Production target: approved origin list and authenticated route access	Open gap

Test Evidence and Sign Off

Manju stores test scripts, sample document permissions, screenshots, API responses and defect records in the Quality section of the Notion workspace. A release cannot move from review to approved until the Technical Reviewer signs the test summary and the Procurement Reviewer confirms that labels and human-review instructions are appropriate.

User Manual

Using Reference Cases

Step	User action	What to check
1	Open Assessments and select a reference procurement case.	Read the case description and outcome label.
2	Choose Tender, Bidder pack or Supporting proof in the document reader.	Confirm how a tender clause relates to evidence.
3	Select Prepare assessment.	Watch the workflow move through document reading, requirement extraction and evidence organisation.
4	Review the compliance indicator and evidence register.	Filter Verified, Review or Action rows as needed.
5	Select Download report if a static handoff is required.	Confirm score, reason and evidence reference before use.

Running a Local PDF Assessment

1. Install the dependency list from requirements.txt. 2. Start the FastAPI service on port 8000. 3. Host the project folder on a local web server. 4. Attach a tender PDF and one or more bidder PDFs. 5. Start the compliance review. 6. Review the results with the authorised officer. The workflow supports machine-readable PDFs only in the current prototype.

Interpreting Statuses

Status	Meaning	User response
Verified	The scoped requirement text and supporting evidence were found by the current rule.	Review the cited page before relying on it.
Review	The current rule or evidence requires an officer judgement.	Open the source document and record the officer decision.
Action	The requirement or evidence was not found under the current rule.	Request clarification or treat as a manual evaluation item according to policy.

Release Notes

Release	Date	Included	Known limitations
1.0	22 Sep 2026	Evidence-led browser workspace, four reference cases, tender and bidder PDF upload, FastAPI assessment endpoints, scoped network rules, result filters and report export.	No OCR, no authentication, no restricted CORS policy, no live GeM integration and no legal or award decision automation.

Post Mortem and Lessons Learned

The project remains at prototype stage, so this section records early lessons rather than a final production post-mortem.

Observation	Evidence	Improvement
Evidence references improve review confidence	The results model records source page and reason for each supported check.	Keep evidence mapping mandatory as coverage expands.
Rule scope must remain explicit	The backend currently recognises a limited network equipment rule set.	Publish rule catalogue and test set with each release.
Reference cases improve the demonstration	The interface provides selectable tender and bidder packs without requiring a backend run.	Separate reference cases clearly from live assessment results.
Security controls cannot be implied	The prototype permits all CORS origins and lacks user authentication.	Treat security backlog as a release gate for any external deployment.

Maintenance and Support Guide

Support area	Runbook action	Escalation
Service unavailable	Check FastAPI process, local port 8000, application logs and browser API URL.	Technical Reviewer if service does not recover in 30 minutes
PDF extraction failure	Confirm the file opens locally, is machine-readable and is not password protected. Do not email confidential files outside approved storage.	Backend contributor for reproducible approved test case
Incorrect status	Compare tender clause, bidder excerpt and evidence page. Mark manual review and capture the rule version.	Procurement Reviewer and Technical Reviewer
Security concern	Stop external sharing, record the incident, preserve minimum necessary logs and notify the project owner.	Manju then sponsor security contact
Rule update	Follow the rule-change SOP and regression test before release.	Manju controls release approval

Support Service Targets

For a pilot, acknowledge a documented issue within one working day, classify its severity within two working days and provide a resolution target after reproduction. Production service-level targets require agreement with the deploying organisation and must cover availability, retention, incident response and escalation contacts.

Official Source Figures

The visual below is an unmodified crop of an official PIB infographic. It reports GeM marketplace statistics as of 28 February 2025, including primary and secondary buyers, order volumes, order values and the MSE share. The visual is included as source evidence rather than redrawn as a new chart [R2].

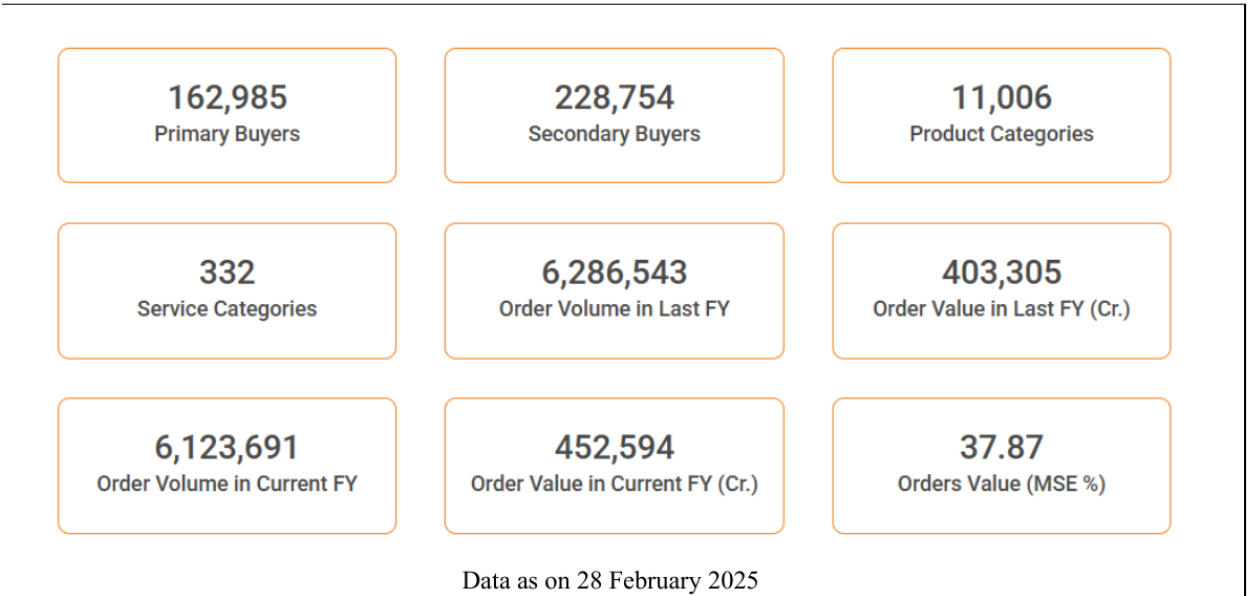


Figure 4. Official GeM statistics visual. Source: PIB, data as on 28 February 2025 [R2].

The official figures demonstrate the scale of the document-review environment. BidWise should not claim that a prototype materially changes these platform-level outcomes. Instead, the pilot should measure its own review time, evidence retrieval time, false-positive rate and officer acceptance rate against a documented baseline.

Interpretation

References

All links below are public, working source locations at the time this document was prepared. They support the stated market facts, figures and technical control recommendations.

R1. Press Information Bureau, Ministry of Commerce and Industry. Government e Marketplace marks 9th Foundation Day and records ₹5.4 lakh crore GMV in FY 2024-25. 8 August 2025. [Open R1 source](#)

R2. Press Information Bureau. Fostering Growth and Inclusivity The Strategic Impact of GeM on India Economy. Official infographic and statistics snapshot, data as on 28 February 2025. [Open R2 source](#)

R3. Press Information Bureau, Ministry of Commerce and Industry. Government e Marketplace surpasses ₹5 lakh crore GMV before FY 2024-25 year-end. 17 March 2025. [Open R3 source](#)

R4. Press Information Bureau. GeM growth journey chart using GeM Transactions and Finance data, data as on 31 March 2023. [Open R4 source](#)

R5. Government e Marketplace official portal and statistics entry point. [Open R5 source](#)

R6. FastAPI official documentation for production API patterns. [Open R6 source](#)

R7. pypdf documentation for PDF text extraction behaviour and limitations. [Open R7 source](#)